

Properties of equality and congruence



1 Complete the following statements.

- a The addition property of equality states that if $a = b$, then $a + c = \text{I}$.
- b The multiplication property of equality states that if $a = b$ and $c \neq 0$, then $ac = \text{I}$.
- c The symmetric property of equality states that if $AB = CD$, then $CD = \text{I}$.

2 By which property of equality are $9x = -27$ and $\frac{9x}{9} = \frac{-27}{9}$ equivalent equations?

3 For each of the following true expressions, determine which property of equality or congruence is illustrated.

- a If $AB = CD$, then $CD = AB$
- b If $AB = CD$, then $AB + 3 = CD + 3$
- c If $\overline{AB} \cong \overline{CD}$ and $\overline{CD} \cong \overline{EF}$, then $\overline{AB} \cong \overline{EF}$

4 If $\overline{PR}$ is a line segment, determine whether the following statements illustrate the reflexive property of congruence.

- | | |
|---------------------------------------|-----------------------------------|
| a $PR \cong RP$ | b $PR = PR$ |
| c $\overline{PR} \cong \overline{PR}$ | d $\overline{PR} = \overline{RP}$ |

5 Consider the following true statement:

$$AB = 12$$

Determine whether the following are valid deductions from this statement.

- | | |
|----------------------|-----------------|
| a $\frac{AB}{2} = 6$ | b $AB + 5 = 17$ |
| c $2AB = 6$ | d $12 = AB$ |

6 Consider the following true statement:

$$AB + CD = 32$$

Determine whether the following is a valid deduction from this statement using only a single property of equality.

- | | |
|-----------------------|-----------------------|
| a $BA + DC = 16 + 16$ | b $AB + AB = 32 + CD$ |
| c $2AB = 64 - CD$ | d $32 = AB + CD$ |



7 For each of the following arguments, determine which property of equality is illustrated.

- a Given: $m\angle ABC = m\angle DEF$ and $m\angle DEF = 75^\circ$
Conclusion: $m\angle ABC = 75^\circ$
- b Given: $PR = QD$
Conclusion: $PR \times PR = PR \times QD$
- c Given: $AB = 16$
Conclusion: $AB - 5 = 11$

8 Consider the following argument.

Given: $\angle TSR \cong \angle QSP$ and $\angle TSR \cong \angle APC$

Conclusion: $\angle QSP \cong \angle APC$

Determine whether the argument relies upon each of the following properties.

- | | |
|--|---|
| a The reflexive property of equality. | b The symmetric property of congruence. |
| c The symmetric property of equality. | d The reflexive property of congruence. |
| e The transitive property of congruence. | f The transitive property of equality. |

Additional questions

9 Copy and complete the table for the properties of equality:

Property of Equality	Example
Reflexive property of equality	$a = a$
Symmetric property of equality	⋮
Transitive property of equality	⋮
Addition property of equality	⋮
⋮	If $a = b$, then $a - c = b - c$.
⋮	If $a = b$, then $a \times c = b \times c$.
Division property of equality	⋮
Substitution property of equality	⋮

- 10 For each of the following pairs of steps, determine which property or converse property of equality is used to get from Step 1 to Step 2.



a

$$\begin{array}{l} \textcircled{1} \quad \angle A = \angle B \\ \textcircled{2} \quad \angle A + \angle C = \angle B + \angle C \end{array}$$

b

$$\begin{array}{l} \textcircled{1} \quad 3x = 24 \\ \textcircled{2} \quad x = 8 \end{array}$$

c

$$\begin{array}{l} \textcircled{1} \quad p + r = q + r \\ \textcircled{2} \quad p = q \end{array}$$

d

$$\begin{array}{l} \textcircled{1} \quad \angle X - \angle Z = \angle Y - \angle Z \\ \textcircled{2} \quad \angle X = \angle Y \end{array}$$

- 11 For each of the following sets of steps, determine which property or converse property of equality is used to get to Step 3.

a

$$\begin{array}{l} \textcircled{1} \quad x = 23 \\ \textcircled{2} \quad y = 23 \\ \textcircled{3} \quad x = y \end{array}$$

b

$$\begin{array}{l} \textcircled{1} \quad x = y + 2 \\ \textcircled{2} \quad 2x - 3y = -4 \\ \textcircled{3} \quad 2(y + 2) - 3y = -4 \end{array}$$